

EX NO: 7

CALCULATION AND VERIFICATION OF FREQUENCY DEVIATION AND MODULATION INDEX OF FM SIGNALS USING

MATLAB SIMULATION

Objectives:

1. To generate a Frequency Modulated (FM) signal in MATLAB using specified carrier and message signal parameters.
2. To calculate and verify the frequency deviation of the FM signal from the given carrier frequency and modulating signal parameters.
3. To determine and verify the modulation index of the FM signal using the relationship between frequency deviation and message frequency.
4. To analyze the effect of message frequency on frequency deviation and modulation index, and verify their relationship through MATLAB simulation

CODE

```
% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signal
clc;
clear;
close all;
%% Message Signal Parameters
Am = 2; % Message amplitude (V)
fm = 500; % Message frequency (Hz)
%% Carrier Signal Parameters
Ac = 1; % Carrier amplitude (V)
fc = 10000; % Carrier frequency (Hz)
%% FM Parameters
kf = 250; % Frequency sensitivity (Hz/V)
fs = 100000; % Sampling frequency (Hz)

T = 0.01; % Simulation duration (s)
%% Time Vector
t = 0:1/fs:T;
%% Message Signal
m = Am*cos(2*pi*fm*t);
%% Frequency Deviation
delta_f = kf*Am;
```

```

%% Modulation Index
beta = delta_f/fm;
%% FM Signal
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
%% Display Parameters
fprintf('Carrier Frequency (fc) = %.0f Hz\n',fc);
fprintf('Message Frequency (fm) = %.0f Hz\n',fm);
fprintf('Message Amplitude (Am) = %.0f V\n',Am);
fprintf('Frequency Sensitivity (kf) = %.0f Hz/V\n',kf);
fprintf('\nFrequency Deviation (Delta f) = %.2f Hz\n',delta_f);
fprintf('Modulation Index (Beta) = %.2f\n',beta);
%% Verification
beta_verify = delta_f/fm;
fprintf('Verified Modulation Index = %.2f\n',beta_verify);
%% Verification Check
tolerance = 1e-6;
if abs(beta - beta_verify) < tolerance
    fprintf('\nVerification Successful\n');
else
    fprintf('\nVerification Failed\n');
end
%% Plotting
figure;
subplot(2,1,1);
plot(t,m,'LineWidth',1.5);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(2,1,2);
plot(t,fm_signal,'LineWidth',1);
grid on;
title('FM Signal');

xlabel('Time (s)');
ylabel('Amplitude (V)');

```

OUTPUT

